

Problem Specification

- **Task:** Matrix Multiplication $C = A \times B$
- **Matrix Size (N):** 32 x 32
- **Data Type:** FP32 (4 bytes per element)
- **Storage:** Row-major order

Task 1: Naive Triple Loop (ijk order)

In the naive implementation, we assume zero data reuse (every access result in a DRAM fetch).

- **Accesses to B per output element:** To compute one element $C[i][j]$, we perform a dot product of a row in A and a column in B. This requires 32 accesses to Matrix B.
- **Total element accesses (A and B):** Matrix A is accessed N times for every element in C (N^2 elements). Total = N^3 .
 - Matrix B is accessed N times for every element in C. Total = N^3 .
 - Total Accesses = $2 \times 32^3 = 65,536$ accesses
- **Total DRAM Traffic:**
 - We must account for reading A and B, and finally writing the result C back to DRAM.
 - Formula: $(N^3 + N^3 + N^2) \times 4$ bytes
 - Calculation: $(32,768 + 32,768 + 1,024) \times 4 = 66,560 \times 4 = 266,240$ Bytes

Task 2: Tiled Loop Analysis (Tile Size T=8)

By blocking the computation into 8 x 8 tiles, we utilize local SRAM (Shared Memory) to reuse data.

- **DRAM loads per tile:** To compute one $T \times T$ tile of C, we load a strip of tiles from A and B.
 - Number of tiles across the dimension = $N/T = 32/8 = 4$.
 - We load 4 tiles of A and 4 tiles of B from DRAM into local memory.
- **Total DRAM Traffic:**
 - With tiling, each element of A and B is loaded from DRAM only N/T times.
 - Formula: $((N^2 \times N/T) + (N^2 \times N/T) + N^2) \times 4$ bytes
 - Calculation: $((1024 \times 4) + (1024 \times 4) + 1024) \times 4 = 9,216 \times 4 = 36,864$ Bytes

Task 3: Traffic Efficiency Ratio

- **Ratio:** Naive Traffic/Tiled Traffic = $266,240/36,864 \approx 7.22$. Note: the theoretical ratio $N/T = 32/8 = 4$ assumes only reads of A and B (excluding the write of C). When C writes are included in both cases, the ratio is approximately 7.22.

- **Justification:** Tiling reduces DRAM traffic by a factor of approximately T (tile size) because each element loaded into high-speed local memory is reused T times to compute an entire block of the output matrix before being discarded.

Task 4: Execution Time and Classification

- **Assumed Specs:** Bandwidth = 320 GB/s, Peak Compute = 10 TFLOPS.
- **Total Operations:** $2 \times N^3 = 65,536$ FLOPs
- **Naive Execution Time:** $266,240 \text{ bytes} / (320 \times 10^9 \text{ bytes/s}) = 0.832$ microseconds.
 - **Bound: Memory-Bound.** The low arithmetic intensity forces the kernel to be limited by the bandwidth roof.
- **Tiled Execution Time:** $36,864 \text{ bytes} / (320 \times 10^9 \text{ bytes/s}) = 0.115$ microseconds.
 - **Bound: Compute-Bound** (or much closer to the ridge). By reducing memory traffic, the bottleneck shifts from the memory bus toward the actual processing units.